

Индивидуальный проект №4

Презентация

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1 Информация

2 Выполнение лабораторной работы

Раздел 1

Информация

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Раздел 2

Выполнение лабораторной работы

getenforce и sestatus.

```
iazubov@iazubov:~$ getenforce
```

```
Enforcing
```

```
iazubov@iazubov:~$ sestatus
```

```
SELinux status:                enabled
SELinuxfs mount:                /sys/fs/selinux
SELinux root directory:         /etc/selinux
Loaded policy name:              targeted
Current mode:                    enforcing
Mode from config file:           enforcing
Policy MLS status:               enabled
Policy deny_unknown status:      allowed
Memory protection checking:      actual (secure)
Max kernel policy version:       33
```

⋮

Обратимся с помощью браузера к веб-серверу

```
iazubov@iazubov:~$ sudo systemctl start httpd
iazubov@iazubov:~$ sudo systemctl enable httpd
Created symlink '/etc/systemd/system/multi-user.target' to
```

```
iazubov@iazubov:~$ sudo systemctl status httpd
```

● httpd.service - The Apache HTTP Server

Loaded: loaded (/usr/lib/systemd/system/httpd.service; **enabled**; preset: **disabled**)

Active: **active (running)** since Thu 2026-04-30 20:58:29 MSK; 20s ago

Invocation: 61c94465a300430a8d19299f10f001d2

Docs: man:httpd.service(8)

Main PID: 6954 (httpd)

Status: **"Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec: 0 B/sec"**

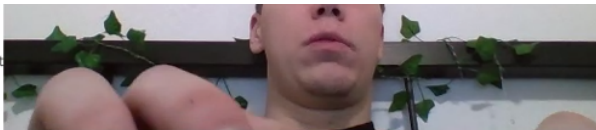
Tasks: 177 (limit: 23028)

Memory: 17.9M (peak: 18.5M)

CPU: 349ms

CGroup: /system.slice/httpd.service

```
├─6954 /usr/sbin/httpd -DFOREGROUND
├─6955 /usr/sbin/httpd -DFOREGROUND
├─6956 /usr/sbin/httpd -DFOREGROUND
├─6957 /usr/sbin/httpd -DFOREGROUND
└─6958 /usr/sbin/httpd -DFOREGROUND
```



```
anp 30 20:58:28 iazubov.localdomain systemd[1]: Starting httpd.service - The Apache HTTP Server...
```

```
anp 30 20:58:28 iazubov.localdomain (httpd)[6954]: httpd.service: Referenced but unset environment variable evaluated
```

```
anp 30 20:58:29 iazubov.localdomain httpd[6954]: Server configured, listening on: port 80
```

```
anp 30 20:58:29 iazubov.localdomain systemd[1]: Started httpd.service - The Apache HTTP Server.
```

Найдем веб-сервер Apache и информацию

```
iazubov@iazubov:~$ ps auxZ | grep httpd
system_u:system_r:httpd_t:s0    root      6954  0.2  0.2  19136 11148 ?        Ss   20:58   0:00 /usr/sbin/httpd -D
FOREGROUND
system_u:system_r:httpd_t:s0    apache    6955  0.0  0.1  18720  5308 ?        S    20:58   0:00 /usr/sbin/httpd -D
FOREGROUND
system_u:system_r:httpd_t:s0    apache    6956  0.0  0.2  2354468 8156 ?        Sl   20:58   0:00 /usr/sbin/httpd -D
FOREGROUND
system_u:system_r:httpd_t:s0    apache    6957  0.0  0.2  2157796 7924 ?        Sl   20:58   0:00 /usr/sbin/httpd -D
FOREGROUND
system_u:system_r:httpd_t:s0    apache    6958  0.0  0.2  2157796 7924 ?        Sl   20:58   0:00 /usr/sbin/httpd -D
FOREGROUND
unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 iazubov  7671  0.0  0.0  227712 2328 pts/0    S+   21:00   0:00 grep --
color=auto httpd
iazubov@iazubov:~$ sestatus -bigrep httpd
sestatus: invalid option -- 'i'
```

Usage: sestatus [OPTION]

- v Verbose check of process and file contexts.
- b Display current state of booleans.

Without options, show SELinux status.

```
iazubov@iazubov:~$ seinfo
bash: seinfo: команда не найдена...
```

Определяем типы файлов и создаем файл

```
iazubov@iazubov:~$ ls -lZ /var/www
```

итого 0

```
drwxr-xr-x. 2 root root system_u:object_r:httpd_sys_script_exec_t:s0 6 дек 10 03:00 cgi-bin
```

```
drwxr-xr-x. 2 root root system_u:object_r:httpd_sys_content_t:s0      6 дек 10 03:00 html
```

```
iazubov@iazubov:~$ ls -ld /var/www/html
```

```
drwxr-xr-x. 2 root root 6 дек 10 03:00 /var/www/html
```

```
iazubov@iazubov:~$ sudo touch /var/www/html/test.html
```

```
[sudo] пароль для iazubov:
```

```
iazubov@iazubov:~$ sudo nan /var/www/html/test.html
```

...

Проверка



```
iazubov@iazubov:~$ chcon -t samba_share_t /var/www/html/test.html
chcon: не удалось изменить контекст безопасности '/var/www/html/test.html' на «unconfined_u:object_r:samba_share_t:s0»: Операция не позволена
iazubov@iazubov:~$ sudo chcon -t samba_share_t /var/www/html/test.html
iazubov@iazubov:~$ ls -Z /var/www/html/test.html
```

...

Запустим веб-сервер Apache на прослушивание TCP-порта 81

```
GNU nano 8.1 /etc/httpd.conf
# See the httpd.conf(5) man page for more information
# and httpd.service(8) on using and configuring the ht
#
# Do NOT simply read the instructions in here without
# what they do. They're here only as hints or reminders
# consult the online docs. You have been warned.
#
# Configuration and logfile names: If the filenames you
# of the server's control files begin with "/" (or "d"
# server will use that explicit path. If the filename
# with "/", the value of ServerRoot is prepended -- so
# with ServerRoot set to '/www' will be interpreted by
# server as '/www/log/access_log', whereas as '/log/acce
# interpreted as '/log/access_log'.
#
# ServerRoot: The top of the directory tree under which
# configuration, error, and log files are kept.
#
# Do not add a slash at the end of the directory path.
# ServerRoot at a non-local disk, be sure to specify a
# Mutex directive, if file-based mutexes are used. If
# same ServerRoot for multiple httpd daemons, you will
# least PidFile.
#
ServerRoot "/etc/httpd"
#
# Listen: Allows you to bind Apache to specific IP addre
# ports, instead of the default. See also the <Virtual
# directive.
#
# Change this to Listen on a specific IP address, but
# httpd.service is enabled to run at boot time, the ac
# available when the service starts. See the httpd.se
```

список портов и контекст

```
iazubov@iazubov:~$ sudo tail -n 1 /var/log/messages
Apr 30 21:25:19 iazubov systemd[13236]: Reached target timers.target - Timers.
iazubov@iazubov:~$ sudo semanage port -a -t http_port_t -p tcp 81
Port tcp/81 already defined, modifying instead
iazubov@iazubov:~$ semanage port -l | grep http_port_t
ValueError: Политика SELinux не задана, или нет доступа к хранилищу.
iazubov@iazubov:~$ sudo semanage port -l | grep http_port_t
http_port_t                tcp      81, 80, 81, 443, 488, 8008, 8009, 8443, 9000
http_port_t                udp      80, 443
pegasus_http_port_t         tcp      5988
iazubov@iazubov:~$ sudo chcon -t httpd_sys_content_t /var/www/html/test.html
iazubov@iazubov:~$
```

...

Проверка сайта (test)

```
iazubov@iazubov:~$ ls -Z /var/www/html/test.html
unconfined_u:object_r:httpd_sys_content_t:s0 /var/www/html/test.html
iazubov@iazubov:~$ curl http://127.0.0.1:81/test.html
curl: (7) Failed to connect to 127.0.0.1 port 81 after 0 ms: Could not connect to server
iazubov@iazubov:~$ echo "Listen 81" | sudo tee -a /etc/httpd/conf/httpd.conf
[sudo] пароль для iazubov:
Попробуйте ещё раз.
[sudo] пароль для iazubov:
Listen 81
iazubov@iazubov:~$ sudo grep Listen /etc/httpd/conf/httpd.conf
# Listen: Allows you to bind Apache to specific IP addresses and/or
# Change this to Listen on a specific IP address, but note that if
#Listen 12.34.56.78:80
Listen 80
Listen 81
iazubov@iazubov:~$ curl http://127.0.0.1:81/test.html
curl: (7) Failed to connect to 127.0.0.1 port 81 after 0 ms: Could not connect to server
iazubov@iazubov:~$ sudo systemctl restart httpd
iazubov@iazubov:~$ sudo ss -tlnp | grep :81
LISTEN 0      511          *:81          *:~      users:(("httpd",pid=15185,fd=6),("httpd",pid=15135,fd=6),("
httpd",pid=15133,fd=6),("httpd",pid=15128,fd=6))
iazubov@iazubov:~$ curl http://127.0.0.1:81/test.html
<html>
```


Удаляем привязку http_port_t к 81 порту. Удаляем файл /var/www/html/test.html

```
iazubov@iazubov:~$ sudo semanage port -d -t http_port_t -p tcp 81
iazubov@iazubov:~$ rm /var/www/html/test.html
```

```
rm: удалить защищённый от записи обычный файл '/var/www/html/test.html'? y
rm: невозможно удалить /var/www/html/test.html : Отказано в доступе
iazubov@iazubov:~$ sudo rm /var/www/html/test.html
iazubov@iazubov:~$
```

...